

Computer Systems

COMP1071: Durham University

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Contents

Part I

Digital Electronics

1 Transistors and Logic Gates

1.1 Transistors

Definition 1.1.1: Transistors are electronically controlled switches that turn on or off when voltage or current is applied to a control terminal.

Definition 1.1.2: An nMOS transistor is a sandwich of layers of conducting and insulating material. The gate is a conducting layer, on top of an insulating layer, on top of the substrate which is a semi-conductor.

Figure 1:

1.2 Logic Gates

Part II

Machine Architecture

Part III

Databases

Part IV

Operating Systems

Part V

References

2 Module Professors

The below is accurate as of the 2025/26 academic year and will not be updated in subsequent years.

- *Digital Electronics* and *Machine Architecture* are taught by Ioannis Ivrissimtzis.
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3 Cited Works